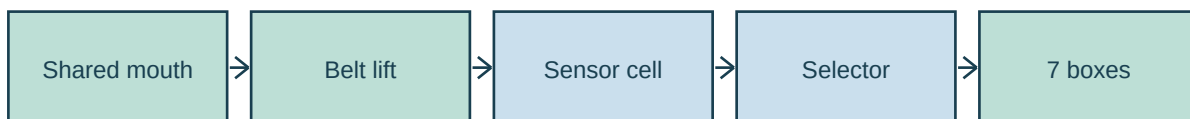


CareBot and SealBot

Simple Arduino architecture and purchasing plan

CareBot: Uno R3, PCA9685, and ESP32-CAM. SealBot: one Uno R3. Shared kit-and-patient lift with seven tilted boxes. Raised-flap beam grippers. JCB disc loader and indexed belt conveyor.



Kits and patients use one lift, one reading cell, and one selector.



Every box is tilted and has its own simple servo outlet gate for batch dumping.

A complete mechanical, electrical, firmware, parts, pricing, dimensions, and prototype plan for the existing two-robot school project.

Status: engineering proposal. Clearances, slopes, sensor thresholds and mission timing require tests with the real pieces and field.

Complete new-build planning total: Rs 16,750, including shipping and repair reserve.

This is the team's architecture, kept simple. CareBot picks up a kit or coloured piece, lifts it to a colour sensor, drops it into the correct tilted box, and later empties that box with a servo gate. SealBot has two beam grippers at the lower front, with the movable input flap directly above them. It places the beams first, then uses that flap and a small belt-and-tray conveyor to collect and release the discs. CareBot uses one Uno R3, one PCA9685, and one ESP32-CAM. SealBot uses one Uno R3 by itself.

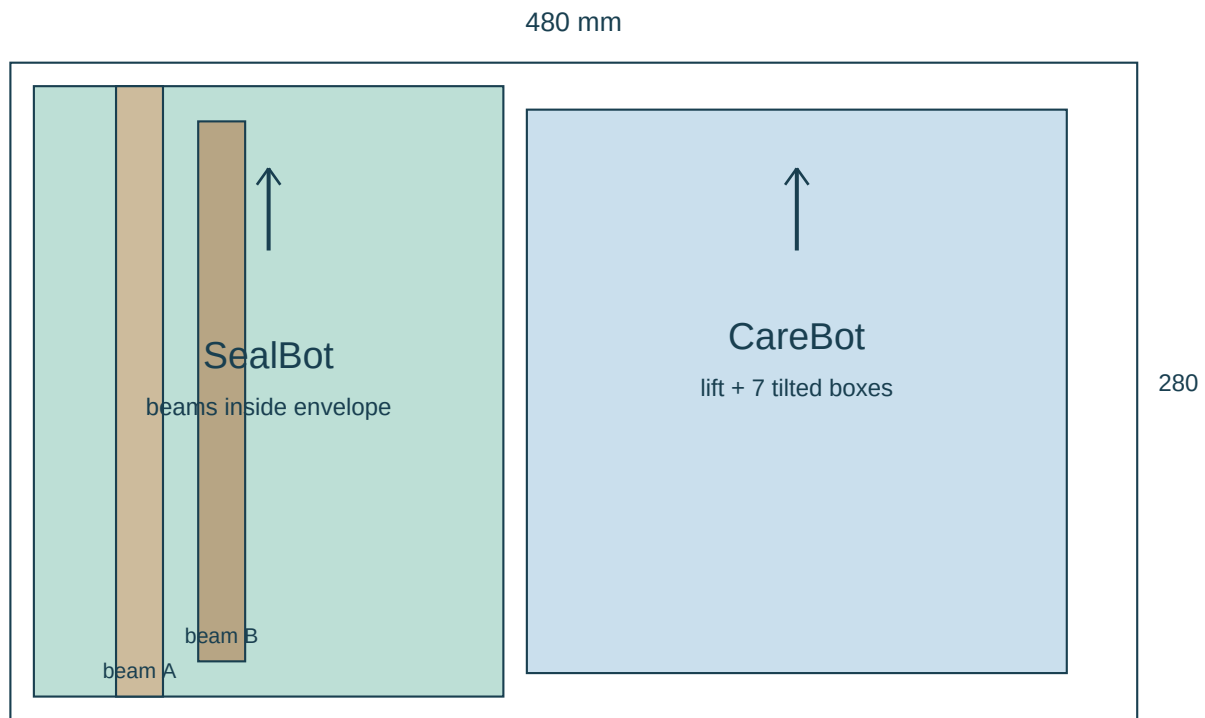
1. Complete architecture

Robot	Front	Inside	Rear
CareBot	One V-shaped pickup mouth and two rubber lift belts	Colour sensor, one servo sorting flap, tilted storage boxes	One simple servo gate on each box
SealBot	Two beam grippers at the bottom, JCB input flap directly above them	One rubber-belt conveyor with three disc trays	Fixed output flap with one servo gate and a line sensor

The Uno R3 is the main controller on both robots. CareBot's PCA9685 produces the nine servo signals, which leaves the Uno free for the motors and sensors. Its ESP32-CAM finds targets and sends short commands such as LEFT, RIGHT, CENTRE, and TARGET FOUND to the Uno. SealBot does not need a camera for the fixed-layout Junior field; its Uno follows the line and runs its four servos directly.

Size lock

Robot	Maximum build envelope	What is included
CareBot	230 W x 240 L x 280 H	Wheels, lift, camera, seven tilted boxes, gates, and servo horns
SealBot	200 W x 260 L x 210 H	Wheels, front grippers, raised input flap, conveyor, rear outlet, and servo horns
Both in the start zone	440 W x 260 L	Includes a 10 mm gap between the robots



Top view. Pair envelope is 440 x 260 mm including a 10 mm gap. Gauge the loaded beams physically.

Figure 1. Proposed starting arrangement, dimensioned in start-local coordinates.

The rulebook starting zone is 480 x 280 mm, so the pair has 40 mm of spare width and 20 mm of spare length. The loaded beams must lie along the 280 mm direction and must be checked in a physical 480 x 280 mm gauge before the final frame is cut.

2. CareBot

Simple pickup

Use two vertical rubber belts facing each other at the front. Each belt runs around two ordinary pulleys. One N20 geared motor turns both belts inward and upward. A V-shaped mouth guides a kit or coloured piece between them, and the belts carry it to the top.

Use wide rubber bands, timing belts, or rubber strips joined into loops. One belt mount can move slightly against a spring so both piece sizes fit. This is the simple replacement for caterpillar wheels.

Sensor and boxes

At the top, the piece stops in front of a TCS34725 colour sensor. Directly below the sensor, one servo flap points toward the required storage box. A second servo opens the small floor under the piece, so it falls into that box.

Use seven tilted boxes because the game pieces have seven final groups:

- Three kit boxes hold 6 for H, 2 for PCC-A, and 2 for PCC-B.
- Four coloured-piece boxes hold red for H, green for RZ, yellow for PCC-A, and yellow for PCC-B.

The kits can use a simple count. The first six go into the H box, the next two into PCC-A, and the final two into PCC-B. The sensor sorts the coloured pieces. Every box is already tilted toward its outlet and has one SG90 servo gate.

CareBot sequence

1. **Pick up one piece.** The camera centres the front mouth, then the rubber belts pull the piece upward.

2. **Read it.** The lift stops at the colour sensor.
3. **Select its box.** One servo points the sorting flap toward the correct box.
4. **Store it.** The floor servo opens and the piece falls into the tilted box.
5. **Dump the group.** At the destination, the camera or line sensor confirms position and the box gate opens. Gravity drops the complete group into the area.

That is the whole CareBot mechanism. There is one lift, one sensor point, one sorting flap, and tilted boxes with gates.

3. SealBot

Exact front layout

The two grippers sit at the lowest front portion of the robot. They hold the two wooden beams standing on the field. The movable input flap sits directly above the grippers. The output flap stays fixed at the rear.

The input flap remains raised during the complete beam task, so it does not hit the beams or block the grippers.

Beam sequence

1. Both beams start standing in the two front grippers.
2. SealBot drives to the first beam area and opens the first gripper.
3. SealBot drives to the second beam area and opens the second gripper.
4. Both beams remain standing in their areas. The beam task is finished.
5. Only now can the input flap move down.

There is no later beam push, no staging step, and no beam mechanism inside the robot.

Disc input flap

The input flap moves like a small JCB bucket. A metal-gear servo lowers it to the field, the robot drives forward until a disc sits on the flap, and the servo raises the flap. At the top, the flap tips the disc onto the first tray inside the robot.

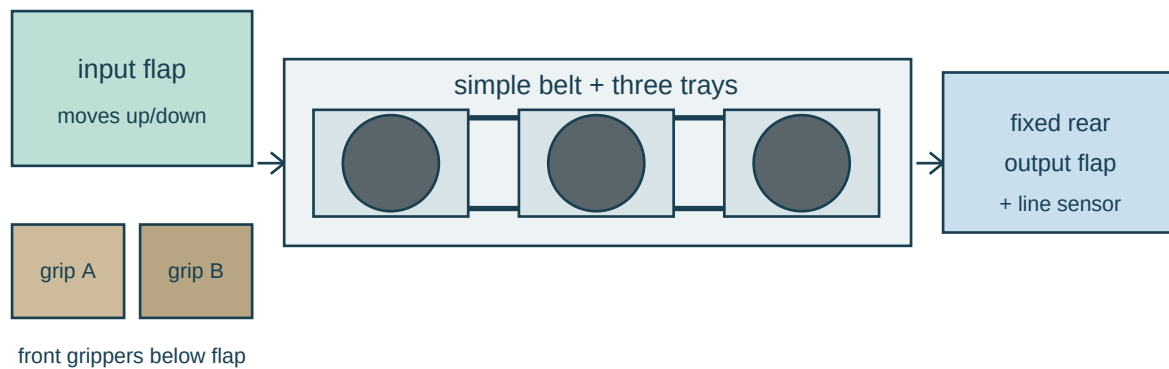
Simple conveyor without caterpillar wheels

Use one homemade loop conveyor. Two rubber belts or timing belts run around a front shaft and a rear shaft. Attach three small flat trays across the belts. One N20 geared motor turns the rear shaft.

After the flap loads a disc, the motor moves the belt forward by one tray position. The next empty tray arrives below the input flap, while the loaded disc moves toward the rear. A small IR sensor or microswitch marks each stopping position.

This gives the same action as the proposed caterpillar wheel and tray, but it only needs ordinary pulleys, rubber belts, a motor, and three flat trays.

SealBot side layout, front at left



Beam phase: flap stays up. Disc phase: flap loads the front tray and the belt moves each disc to the rear.

Figure 2. The grippers are at the lower front, the movable input flap is directly above them, and the output flap stays fixed at the rear.

Rear output

The rear output flap stays in one place. Connect only the left, centre, and right outputs of a downward five-channel line-sensor board. The Uno follows the line to the required mark, then a wheel-encoder distance or a separate reflective marker gives the final stop.

When the correct position is detected, one SG90 opens the output gate and one disc drops. The gate closes, then the conveyor moves forward by one tray position. Repeat this at the next two positions.

4. SealBot sequence

1. **Carry the beams.** Keep the input flap raised and hold both beams in the two lower front grippers.
2. **Drop the beams.** Open each gripper at its area. The beam task ends here.
3. **Enable the flap.** Lower the input flap only after both grippers are empty.
4. **Collect a disc.** Scoop it with the flap and tip it onto the front tray.
5. **Move the tray.** Advance the rubber-belt conveyor by one position.
6. **Reach an output point.** Follow the field line with the left, centre, and right sensor outputs, then stop at the marker or stored encoder count.
7. **Drop one disc.** Open the fixed rear output gate, close it, then advance the next tray.

5. Simple split control

CareBot uses an Uno R3 with one PCA9685 because it has nine servos. SealBot has only four servos, so its Uno R3 drives them directly and needs no expansion board.

Part	Job
CareBot Uno R3	Runs the sequence, three DC motors, colour sensor, switches, and PCA9685
CareBot PCA9685	Produces all nine CareBot servo signals over the two-wire I2C connection
CareBot ESP32-CAM	Finds objects and alignment marks, then sends a short result to the Uno
SealBot Uno R3	Runs three DC motors, four servos, three line-sensor channels, start input, and conveyor index sensor
Two DRV8833 boards	One drives the left and right wheels; the other drives the lift or conveyor motor
TCS34725	Reads the CareBot piece colour at the top of the lift
Five-channel line sensor	SealBot uses only its left, centre, and right outputs for simple line following

The ESP32-CAM remains CareBot's main vision sensor, but it only sends horizontal error, target type, and target visible or lost. The CareBot Uno handles real-time movement. The PCA9685 and TCS34725 share the same A4/A5 I2C connection.

Simple split control

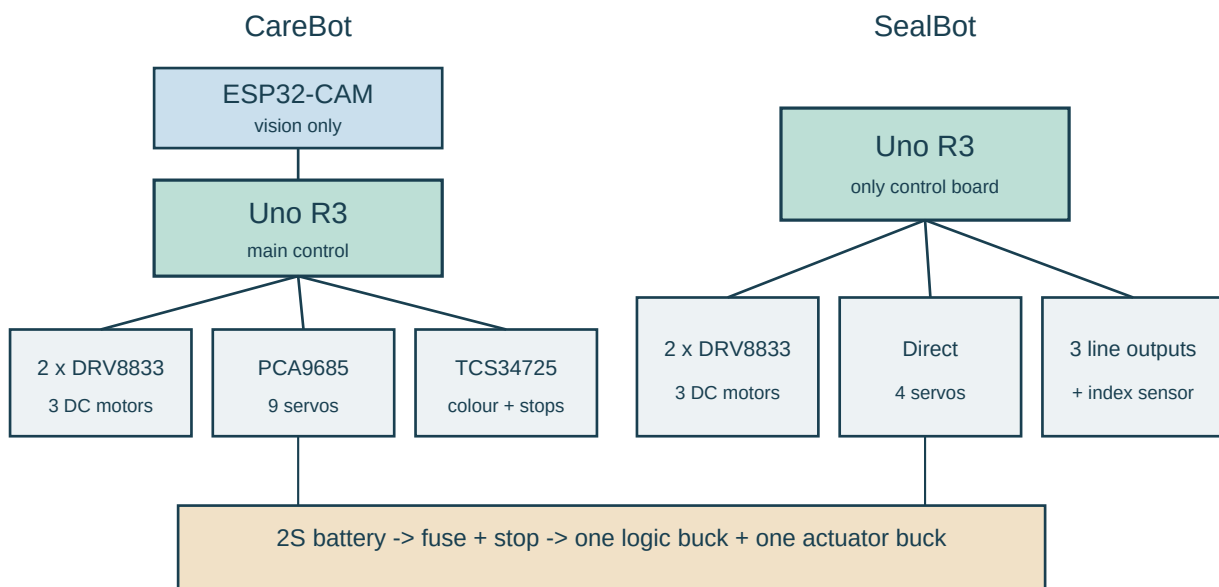


Figure 3. CareBot adds one servo driver and one vision board; SealBot needs only its Uno R3.

SealBot's pin budget fits one Uno without making the wiring awkward. Use D2-D7 for the three DC motors, D8-D11 for the four servos, A0-A2 for the three line signals, A3 for start, and A4 for the conveyor index sensor. D0/D1 remain free for USB serial, and A5 remains spare. The emergency stop cuts actuator power directly and does not need a software pin.

Why the earlier design had many boards

The ESP32-CAM uses most of its pins for the camera, so it cannot control the complete robot reliably by itself. CareBot still needs one PCA9685 because nine servo signals would crowd the Uno. SealBot needs neither a PCA9685 nor a PCF8575 because four direct servo connections and three line signals fit its Uno.

No separate USB-UART programmer is purchased. During setup, hold the CareBot Uno's ATmega328P in reset and use the Uno's onboard USB-to-serial interface to upload the ESP32-CAM firmware. Add a two-resistor divider between the Uno TX and the 3.3 V ESP32-CAM RX. After programming, reconnect the camera to the Uno using two software-serial pins.

Power kept simple

Use two regulators on each robot. One LM2596 provides clean 5 V power for the Uno and, on CareBot, the camera. One XL4015 provides the actuator rail for the servos and motors. Connect all grounds at one common point, and never power the servos from the Uno's 5 V pin. The program moves only the required gate or flap, so one 5 A actuator regulator is enough per robot.

6. Actuator count

Robot	DC motors	Servos
CareBot	2 drive motors and 1 lift motor	1 sensor-floor servo, 1 sorting-flap servo, and 7 box-gate servos
SealBot	2 drive motors and 1 conveyor motor	2 beam-gripper servos, 1 JCB-flap servo, and 1 output-gate servo

The complete build uses thirteen working servos. Buy three spare SG90 servos and one spare MG90S because small servos are easy to damage during testing.

7. Complete parts and pricing

These are planning prices checked on 7 September 2026. Listed prices can change, and stock status must be checked again before payment.

Supplier comparison for the main parts

Part	Quantity	Robocraze Rs	Quartz Rs	Robu Rs
ESP32-CAM with camera	1	689	675	739
Uno R3 CH340 compatible board	2	281	218	229
PCA9685 16-channel servo driver	1	255	212	219
DRV8833 motor-driver board	4	67	63, out of stock	399 current listing
2WD chassis kit	2	285	265	428
SG90 servo	13	91	86	93
MG90S servo	4	149, out of stock	124	125
TCS34725 colour sensor	1	306	191, out of stock	269
Five-channel line sensor	1	189	160	177 listing
LM2596 regulator	2	48	107 with display	41
XL4015 regulator	2	84	79	71
B3 balance charger	1	329	294	336
7.4 V 2200 mAh high-discharge LiPo	2	No close match found	1,221, out of stock	1,399

Quartz currently has the lowest Uno R3 and PCA9685 prices and lists both in stock. The chosen cart still mixes suppliers because Robocraze is cheaper for the DRV8833 and Robu has the selected colour sensor, geared motors, regulators, and high-discharge batteries. The low-cost Rs 366 Quartz lithium-ion pack is not used because its stated maximum discharge current is only 2 A.

Electronics and motion

Part	Quantity	Unit Rs	Subtotal Rs	Use
Uno R3 CH340 compatible board	2	218	436	One main controller per robot
PCA9685 16-channel servo driver	1	212	212	CareBot's nine servo signals
ESP32-CAM with camera	1	675	675	CareBot vision only
DRV8833 motor-driver board	4	67	268	Two per robot
2WD chassis kit with motors and wheels	2	265	530	One donor base per robot
N20 6 V 100 RPM geared motor	2	289	578	CareBot lift and SealBot conveyor
SG90 servo	13	86	1,118	Ten working and three spare
MG90S metal-gear servo	4	124	496	Three working and one spare
Electronics and motion subtotal			4,313	

Sensors and controls

Part	Quantity	Unit Rs	Subtotal Rs	Use
TCS34725 colour sensor	1	269	269	CareBot sensor point
Five-channel line sensor	1	160	160	SealBot uses three outputs
Wheel encoder sensor	4	60	240	Two per robot
IR or reflective position sensor	6	50	300	Lift, flap, conveyor, and output checks
Microswitch	10	20	200	Grippers, gates, and end stops
Start pushbutton	2	25	50	One per robot
Emergency-stop switch	2	150	300	One per robot
Sensors and controls subtotal			1,519	

Power and wiring

Part	Quantity	Unit Rs	Subtotal Rs	Use
7.4 V 2200 mAh 2S LiPo battery	2	1,399	2,798	One per robot

Part	Quantity	Unit Rs	Subtotal Rs	Use
2S/3S balance charger	1	294	294	Shared charger
LiPo-safe bag	1	350	350	Battery storage
LM2596 regulator	2	41	82	Logic and camera power
XL4015 5 A regulator	2	71	142	One actuator rail per robot
Fuses and holders	2 sets	75	150	One set per robot
Capacitors and motor-noise parts	1 lot	250	250	Both robots
Wire, connectors, perfboard, and standoffs	1 lot	900	900	Both robots
Power and wiring subtotal			4,966	

Body and mechanisms

Material	Allowance Rs	Use
Plywood, acrylic, and PET sheet	900	Boxes, trays, flaps, decks, and covers
Screws, nuts, washers, and spacers	600	Structure
Rods, hinges, springs, and linkage wire	450	Flap and grippers
Rubber belts, pulley material, foam, and rubber strips	500	CareBot lift and SealBot conveyor
Chute liners and flexible outlet pieces	250	Tilted boxes and rear output
Cutting, drilling, or small 3D-printed parts	700	Brackets and pulleys
Practice pieces and field mock-up	600	Testing
Labels and decoration	250	Final presentation
Body and mechanism subtotal	4,250	

Total

Group	Rs
Electronics and motion	4,313
Sensors and controls	1,519
Power and wiring	4,966
Body and mechanisms	4,250
Shipping allowance	900
Repair reserve	800
Complete planning total	16,748

Plan on Rs 16,750 if everything is bought new. If the school already has sheet material, wire, fasteners, and workshop access, expect about Rs 12,500 to Rs 13,000. This total already assumes that SealBot has no camera.

8. What to build next

1. Build one CareBot pickup using four pulleys, two rubber loops, and one N20 motor. Test it with the real kit and coloured piece.
2. Build one SealBot conveyor using two belt loops, two shafts, and three trays. Make it move exactly one tray per command.
3. Make the SealBot front from cardboard. Put the two grippers at the bottom and the raised input flap directly above them.
4. Wire the SealBot Uno to four servos, three line-sensor outputs, and the two DRV8833 boards. Test line following and one-tray indexing before building the final harness.
5. On CareBot, test the Uno, PCA9685, and three servos first, then add the colour sensor and camera message link. Buy the full set only after both mechanism tests work. A small proof set costs about Rs 3,500 to Rs 4,000.

9. Sources

- [Espressif ESP32-CAM camera pin map](#)
- [Arduino Uno R3 hardware specification](#)
- [NXP PCA9685 product page](#)
- [Texas Instruments DRV8833 product page](#)
- [Robocraze component listings](#)
- [Quartz Components listings](#)
- [Robu component listings](#)